

NETWORKS AND COMPLEXITY

Solution 16-4

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 16.4: Degeneracy of the epidemic threshold [Lvl. 2]

Consider the SIS model in the form

$$\dot{x} = f(x, p) = px(1 - x) - x$$

where we have used x instead of I as the variable. Find the epidemic threshold, and then check the genericity conditions of the fold bifurcation $f_p \neq 0$, $f_{xx} \neq 0$.

Solution

We consider the steady state $x^* = 0$. To find the epidemic threshold, our bifurcation point, we consider the Jacobian

$$f_x = p - 2px - 1 \tag{1}$$

To restrict the Jacobian to the steady state, we substitute $x^* = 0$ and then look for the point where the eigenvalue crosses 0, which yields

$$p - 1 = 0 \tag{2}$$

So the invasion threshold is at $p^* = 1$. Now we can evaluate the genericity conditions

$$f_{xx}(x^*, p^*) = -2p|_{x=0, p=1} = -2 \neq 0, \tag{3}$$

$$f_p(x^*, p^*) = x - x^2|_{x=0, p=1} = 0. \tag{4}$$

This shows that the second condition, the so-called *transversality condition* of the bifurcation theorem is violated.